

OCR Further Mathematics
Mechanics - Dimensional Analysis
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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Question		Answer/Indicative content	Marks	Guidance
1	a	$[F] = \text{MLT}^{-2}$ $\left[m v \frac{dv}{dx} \right] = \frac{[m][v][v]}{[x]} = \frac{\text{ML}^2\text{T}^{-2}}{\text{L}} = \text{MLT}^{-2}$	B1 (AO1.1) B1 (AO2.1) [2]	Correctly finding the dimensions of both sides is sufficient for B1B1; an explicit conclusion is not necessary.
	b	Only quantities with the same dimensions can be added (or subtracted) [so $[a^2] = [x^2]$ which means that $[a] = [x]$]	B1 (AO2.4) [1]	
	c	$[k] \text{M}^{-\frac{1}{2}} (\text{L}^2)^{\frac{1}{2}} = \text{LT}^{-1}$ $[k] = \text{M}^{\frac{1}{2}} \text{T}^{-1}$ Alternative solution $v = km^{-\frac{1}{2}} \sqrt{a^2 - x^2} \Rightarrow k = \frac{vm^{\frac{1}{2}}}{\sqrt{a^2 - x^2}}$ so the units of k are $\text{kg}^{\frac{1}{2}} \text{s}^{-1}$ $[k] = \text{M}^{\frac{1}{2}} \text{T}^{-1}$	M1 (AO2.2a) A1 (AO1.1) Alternative solution M1 A1 [2]	Use of formula for v to derive dimensional equation for $[k]$ Alternative solution Use of formula for v to derive units of k .
	d	$\frac{dv}{dx} = km^{-\frac{1}{2}} (-2x) \frac{1}{2} (a^2 - x^2)^{-\frac{1}{2}}$ $\therefore F = mv \frac{dv}{dx}$ $= m \times km^{-\frac{1}{2}} (a^2 - x^2)^{\frac{1}{2}} km^{-\frac{1}{2}} (-2x) \frac{1}{2} (a^2 - x^2)^{-\frac{1}{2}}$ $\therefore F = -k^2 x$	M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [3]	Use of chain rule to differentiate v wrt x Use of formula for F with m , v and their $\frac{dv}{dx}$ substituted in.
		Total	8	



Question		Answer/Indicative content	Marks	Guidance	
2	a	$[r] = L, [m] = M \text{ and } [U] = LT^{-1}$ $[G] = \left[\frac{U^2 r}{m} \right]$ $\therefore [G] = (LT^{-1})^2 LM^{-1} = L^3 M^{-1} T^{-2}$	B1 (AO 2.1) M1 (AO 1.1) A1 (AO 2.2a) [3]	Correct dimensions for other parameters (U, r and m) so (no need for them to be used for this mark to be awarded). Comparing dimensions, realising that 2 is dimensionless and rearranging AG	Could be done by dimensional analysis e.g. $[G] = L^\alpha M^\beta T^\gamma$ and equate indices using $U = \sqrt{\frac{2Gm}{r}}$ oe
	b	$[P] = (MLT^{-2})/T = ML^2 T^{-3}$ Need $LT^{-1} = M^\alpha L^{2\alpha} T^{-3\alpha} M^\beta L^\beta T^{-2\beta} T^\gamma$ M: $\alpha + \beta = 0, L: 1 = 2\alpha + \beta$ $\alpha = 1, \beta = -1$ T: $-1 = -3\alpha - 2\beta + \gamma$ $\gamma = 0$	B1 (AO 3.3) B1 (AO 3.3) M1 (AO 3.4) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 1.1) [6]	Using $P = WD/t$ oe Realising condition for equation to be dimensionally correct and substituting in dimensions. Comparing to obtain equations in α and β Comparing to obtain equation in γ	ft errors in $[P]$ and / or $[W]$ here and in subsequent method marks provided M, L and T appear at least twice on the RHS
	c	Because $\gamma = 0$, the modelled minimum launch speed V does not depend on the time t for which the engines operate...	B1ft (AO 3.5a) [1]	ie the modified model predicts that V does not vary when t varies	Or appropriate comment from their result, e.g. if $\gamma = -1$, then V is inversely proportional to t
		Total	10		



Question		Answer/Indicative content	Marks	Guidance	
3	a	$[Force] = MLT^{-2}$ $[Viscosity] = MLT^{-2}L^{-2}T = ML^{-1}T^{-1}$	M1 (AO1.2) A1 (AO1.1) [2]		
	b	No it is not possible since a multiplicative constant does not affect the dimensions of the quantity it is multiplying.	B1 (AO2.4) [1]		Or equivalent statement about dimensional analysis only applying to dimensioned quantities
	c	$MLT^{-2} = (ML^{-1}T^{-1})^{\alpha}L^{\beta}(LT^{-1})^{\gamma}$ From M, $\alpha = 1$ L: $1 = \beta + \gamma - \alpha$, T: $-2 = -\alpha - \gamma$ $\beta = 1, \gamma = 1$	M1 (AO3.3) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	Setting up a dimensional equation (using their [Force] and [Viscosity]) Using their equation to derive 2 equations in β and γ	$MLT^{-2} = M^{\alpha} L^{\beta+\gamma-\alpha} T^{-\alpha-\gamma}$ NB – B1 to allow this from M0
	d	$mg - 6\pi\eta rv = m \frac{dv}{dt}$ $\int dt = \int \frac{1}{g - kv} dv$ $t + c = -\frac{1}{k} \ln(g - kv)$ $t = 0, v = 0 \Rightarrow c = -\frac{1}{k} \ln g$ $t - \frac{1}{k} \ln g = -\frac{1}{k} \ln(g - kv) \Rightarrow -kt = \ln(g - kv) - \ln g$ $\Rightarrow -kt = \ln\left(\frac{g - kv}{g}\right) \Rightarrow e^{-kt} = \frac{g - kv}{g}$ Alternative method 1	M1 (AO3.3) A1 (AO1.1) M1 (AO1.1) M1 (AO3.4) A1 (AO1.1) M1	Using $F = ma$ with two forces Correct Correctly separating variables in v and t Integrating and substituting initial values to find c AG. Must see some intermediate working Using $F = ma$ with two forces	any differential expression for a no need for k to be substituted yet



Question			Answer/Indicative content	Marks	Guidance	
			$mg - 6\pi\eta r v = m \frac{dv}{dt}$ $g = kv + \frac{dv}{dt} \text{ so Integrating Factor } e^{\int k dt} = e^{kt}$ $ge^{kt} = kve^{kt} + \frac{dv}{dt}e^{kt} = \frac{d}{dt}(ve^{kt})$ $ve^{kt} = \int ge^{kt} dt = \frac{g}{k}e^{kt} + c$ $t = 0, v = 0 \Rightarrow c = -\frac{g}{k}$ $ve^{kt} = \frac{g}{k}e^{kt} - \frac{g}{k} \Rightarrow e^{kt}\left(\frac{g}{k} - v\right) = \frac{g}{k}$ $\Rightarrow e^{-kt} = \frac{\frac{g}{k} - v}{\frac{g}{k}} = \frac{g - kv}{g}$ Alternative method 2 $mg - 6\pi\eta r v = m \frac{dv}{dt}$ $g = kv + \frac{dv}{dt}$ CF: $\frac{dv}{dt} + kv = 0 \Rightarrow \frac{dv}{dt} = -kv \Rightarrow \int \frac{1}{v} dv = -\int k dt$ $\Rightarrow \ln v = -kt + c \Rightarrow v = Ae^{-kt}$ PI: try $v = \alpha \Rightarrow k\alpha = g \Rightarrow \alpha = \frac{g}{k}$, so GS: $v = Ae^{-kt} + \frac{g}{k}$ $t = 0, v = 0 \Rightarrow 0 = A + \frac{g}{k} \Rightarrow A = -\frac{g}{k}$ $\text{So } v = -\frac{g}{k}e^{-kt} + \frac{g}{k} \Rightarrow e^{-kt} = \frac{k}{g}\left(\frac{g}{k} - v\right) = \frac{g - kv}{g}$	A1 M1 M1 A1 M1 A1 M1 A1 [5]	Rearrange and find IF Substituting initial values to find c AG. Must see some intermediate working Using $F = ma$ with two forces Rearrange and find CF and PI Substituting initial values to find c AG. Must see some intermediate working	any differential expression for a no need for k to be substituted yet any differential expression for a no need for k to be substituted yet
	e		$e^{-kt} = \frac{g - kv}{g} \text{ so } t \rightarrow \infty, \Rightarrow v_T = \frac{g}{k}$ Alternative method $\frac{dv}{dt} = 0 \Rightarrow mg - 6\pi\eta r v_T = 0 \Rightarrow v_T = \frac{mg}{6\pi\eta r} = \frac{g}{k}$	B1 (AO3.4) B1 [1]		



Question			Answer/Indicative content	Marks	Guidance	
	f		As the viscosity increases the terminal velocity decreases... ...and as the viscosity tends to infinity the terminal velocity tends to 0	B1 (AO2.2a) B1 (AO2.2a) [2]		
			Total	15		
4	a		$[v^2] = [pu^\alpha]$ and $[v] = [u]$ and p is dimensionless or $L^2T^{-2} = L^\alpha T^{-\alpha} \Rightarrow \alpha = 2$	M1 (AO2.1) A1 (AO2.1) [2]	For the idea, that might be stated in words, that the dimensions of every term in the equation must be the same AG	Must relate LHS to RHS Must explicitly state that p is dimensionless to get this mark (this may be seen in part (b)).
	b		$L^2T^{-2} = (LT^{-2})^\beta L^\gamma$ $-2 = -2\beta \Rightarrow \beta = 1$ $\beta + \gamma = 2$ $\gamma = 1$	M1 (AO3.3) A1 (AO3.3) M1 (AO1.1) A1 (AO1.1) [4]	Equating dimensions of other term with $[v^2]$ with q gone and $[a]$ and $[s]$ used Ignore term in $[u^2]$ if present Equating powers of L	May be seen expanded ASM note – must be seen or strongly implied – see wording of question. SC2 for both correct
	c		If $s = 0$ then $v = u$ so $u^2 = pu^2 + 0 \Rightarrow p = 1$	B1 (AO3.4) [1]	Details must be shown	Do not allow use of prior knowledge of $v^2 = u^2 + 2s$.
	d		$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}qmas = \frac{1}{2}qFs = \frac{1}{2}qW$ So we can see that this equation is a statement of the Work-Energy principle so $\frac{1}{2}q = 1$ so $q = 2$	M1 (AO3.4) A1 (AO2.4) [2]	Where F must be the force acting and W the work done by this force	Do not allow use of prior knowledge of $v^2 = u^2 + 2s$.
			Total	9		



Question		Answer/Indicative content	Marks	Guidance	
5	a	$[a] = LT^{-2}$ $[h] = L$ and $[m] = M$ and $[v] = LT^{-1}$ $M: 0 = 1 + \alpha$ so $\alpha = -1$ $L: 1 = 1 + \beta + \gamma$ $T: -2 = -1 - \beta$ $\beta = 1$ and $\gamma = -1$	B1(AO3.3) B1(AO1.2) B1(AO3.3) M1(AO1.1) M1(AO3.4) A1(AO1.1) [6]	www	<u>Examiner's Comments</u> This was the question with the highest success rate in this examination. The most common error was using units of M as part of the acceleration. Examiners would encourage candidates to show which dimension they are equating to make sure they gain method marks if they make a mistake.
	b	$(9.11 \times 10^{-31})^{-1}$ or $(1.67 \times 10^{-27})^{-1}$ oe $(9.11 \times 10^{-31})^{-1} : (1.67 \times 10^{-27})^{-1}$ $1.67 \times 10^{-27} : 9.11 \times 10^{-31}$ or awrt 1830 (:1)	M1(AO3.4) M1(AO2.2a) A1(AO1.1) [3]	Using $a \propto \frac{1}{m}$ Division of values or forming a ratio Accept answers in words (eg the acceleration of the electron is 1830 times the acceleration of the proton)	Could be either way round Reciprocals not acceptable in final answer <u>Examiner's Comments</u> This was generally well answered. In some cases candidates found it challenging to logically express the comparisons and some did not denote which side related to electron or proton. This made it difficult to award method marks. A common error was giving the ratio the wrong way round.



Question			Answer/Indicative content	Marks	Guidance
	c		Because N is a dimensionless quantity	E1 (AO3.5b) [1]	Must see 'dimensionless' or 'no dimensions' <u>Examiner's Comments</u> The majority of candidates answered this correctly.  Misconception Referencing 'no units' rather than 'dimensionless'
			Total	10	



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Question			Answer/Indicative content	Marks	Guidance
6	a		$[\rho] = \text{ML}^{-3}$	B1(AO3.3) [1]	If M, L and T not used B0, but do not penalise any further instances of non-standard notation as long as it is used consistently. <u>Examiner's Comments</u> This question required candidates to use the relevant standard dimensional units, i.e. M, L and T, to express the units of density. The vast majority of candidates gained this mark. A few errors and misconceptions were seen, primarily through the use of non-standard units such as <i>m</i>, <i>kg</i> and <i>s</i> instead of the correct ones. This was only penalised once throughout the question.

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Question		Answer/Indicative content	Marks	Guidance	
	b	$[p] = MLT^{-2}L^{-2} = ML^{-1}T^{-2}$ $LT^{-1} = M^{\alpha}L^{-2\alpha}M^{\beta}L^{-3\beta}L^{\gamma}$ M: $\alpha + \beta = 0$ T: $-2\alpha = -1$ $\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$ L: $1 = -\alpha - 3\beta + \gamma$ $\gamma = 0$ www	B1(AO2.1) B1ft(AO3.3) M1(AO3.4) M1(AO3.4) A1(AO1.1) M1(AO3.4) A1(AO1.1) [7]	If M, L and T not used B0, provided this has not been withheld in part (a). Award remaining marks where working is clear. Or $(MLT^{-1}T^{-2})^{\alpha}(ML^{-3})^{\beta}(L)^{\gamma}$ ft their expressions for p and p here and in subsequent method marks provided M, L and T present with M and L appearing at least twice on the RHS Ignore use of k in the equation. Allow this mark as long as the equations for M and T are correct. SC2 for three correct values unsupported or SC1 for correct values of α and β or for γ. These may not be combined with any other marks. <u>Examiner's Comments</u> This question required candidates to use the method of dimensional analysis to determine the values of the exponents of a formula. Despite being considerably more challenging than last year's question on the	Do not allow any marks for using addition instead of multiplication

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Question			Answer/Indicative content	Marks	Guidance
					same topic, candidates were much better prepared for this question and about half gained full marks, with most of the rest gaining at least half marks. The first mark was given for finding an expression for the pressure. Most were able to do this successfully, but a significant minority produced the wrong expression, often because they confused the common units with the standard notation. In particular, there were many instances where Force was correctly quoted as MLT^{-2} but then the area was mistakenly written as M^2 (from m^2) rather than L^2, giving $(\text{MLT}^{-2}) \div (\text{M}^2) = \text{M}^{-1} \text{LT}^{-2}$ for the pressure instead of $\text{ML}^{-1} \text{T}^{-2}$. The second mark was for the dimensional equation. Some candidates did not realise that they needed to explicitly substitute their expressions for p, ρ and λ into the given equation $u = k p^\alpha \rho^\beta \lambda^\gamma$, even though the question asks them to determine the values of the exponents. Similarly, a very small number of candidates did not produce the required equations for each of the dimensions to determine the values of α, β and γ and as a result were not able to gain any credit for their answers.

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Question		Answer/Indicative content	Marks	Guidance
	c	Sounds of any wavelength have the same speed through the gas	E1FT(AO 2.2b) [1]	Follow from their γ: If $\gamma > 0$ then speed increases as wavelength increases (or better – e.g. $\gamma = \frac{1}{2} \rightarrow$ speed is proportional to $\sqrt{\lambda}$); if $\gamma < 0$ then speed decreases as wavelength increases (or better) <u>Examiner's Comments</u> This was another question requiring interpretation of the values found in a previous part of a question, in this case using the candidate's value of γ to determine the effect of wavelength on the speed of the wave. Most candidates correctly interpreted their values. Some made statements that were incorrect or too vague to gain any credit, e.g. (for $\gamma = 3$): "the wavelength ... makes a considerable difference to the speed if it is changed", without saying what kind of difference it would make. Others used inappropriate terms such as distance instead of wavelength, or talked about the value of γ instead of λ, which again did not gain any credit.

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Question			Answer/Indicative content	Marks	Guidance
	d		$u \propto \frac{1}{\sqrt{\rho}}$ or $u = k\sqrt{\frac{p}{\rho}}$ oe $\frac{1}{4}$	M1(AO3.4) A1(AO1.1) [2]	$2 = \left(\frac{\rho_B}{\rho_A}\right)^{-\frac{1}{2}}$ Award if no working seen provided $\beta = -\frac{1}{2}$ <u>Examiner's Comments</u> In this question, candidates were required to use their value of β to quantify the change in density required to make the wave travel at twice the original speed. Most candidates attempted this question and some were successful. Whilst most showed some awareness of the formula by substituting their value of β , not all were able to use the mathematical concept of proportionality. In this case, $v \propto \frac{1}{\sqrt{\rho}}$, meaning that $\rho \propto \frac{1}{v^2}$. This gave an answer of $\frac{1}{4}$, assuming that β was correct.
			Total	11	



Question			Answer/Indicative content	Marks	Guidance	
7		a	$[F] = [ma] \text{ MLT}^{-2}$ $[P] = \frac{[F]}{[A]} = \text{MLT}^{-2}\text{L}^{-2} = \text{ML}^{-1}\text{T}^{-2}$ $[R] = [F][d]$ $[R] = \text{MLT}^{-2}\text{L} = \text{ML}^2\text{T}^{-2}$	M1 (AO 1.2) A1 (AO 2.2a) M1 (AO 3.1b) A1 (AO 3.2a) [4]	Determining dimensions of energy	
		b	$(\text{ML}^{-1}\text{T}^{-2})^\alpha \text{L}^{3\beta} = \text{ML}^2\text{T}^{-2}$ $\text{M}^\alpha = \text{M} \Rightarrow \alpha = 1 \quad (\text{AG})$ $\text{L: } -\alpha + 3\beta = 2 \Rightarrow \beta = 1 \quad (\text{AG})$	M1 (AO 3.3) A1 (AO 3.4) A1 (AO 1.1) [3]	Using their dimensions or similarly with T	n and θ dimensionless
		c	$P = \frac{5 \times 8.31 \times 300}{0.03}$ 415 500N m ⁻²	M1 (AO 3.4) A1 (AO 1.1) [2]	Correctly substituting values	With their α and β
		d	$[b] = \text{L}^3$ $[\alpha] = \text{ML}^{-1}\text{T}^{-2}(\text{L}^3)^2$ $[\alpha] = \text{ML}^5\text{T}^{-2}$	B1 (AO 2.2a) M1 (AO 2.2a) A1 (AO 1.1) [3]	Using equality of $[a/V^2]$ and $[P]$	With n dimensionless



Question			Answer/Indicative content	Marks	Guidance	
		e	$P = \frac{5 \times 8.31 \times 300}{0.03}$ awrt 414 000 Nm ⁻²	M1 (AO 3.4) A1 (AO 1.1) [2]	Correctly substituting values	With their α and β
		f	$P + \frac{0.14 \times 5^2}{0.03^2} = \frac{5 \times 8.31 \times 300}{0.03 - 5 \times 3.2 \times 10^{-5}}$ A negative value is not a valid value for pressure... ...and so the model is not valid for these values	B1 (AO 3.4) E1 (AO 3.5b) E1 (AO 2.2a) [3]	Ignore units	OR from $V - nb < 0$, so since R, θ, n and V are all positive $P < 0$
			Total	17		



Question			Answer/Indicative content	Marks	Guidance
8		a	$[E] = [Fx] = \text{MLT}^{-2}\text{L}$ $[E] = \text{ML}^2\text{T}^{-2}$ and $[\omega] = \text{T}^{-1}$ so ML^2T^{-2} $^2 = [I]\text{T}^{-2} \Rightarrow [I] = \text{ML}^2$	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 2.2a) [3]	Use of correct equation for, or units of, energy to find dimensions
		b	$I = mr^2 \dots$ $\dots \times k$ where k is dimensionless	B1 (AO 2.2b) B1 (AO 3.3) B1 (AO 3.4) [3]	Must be “I =”
		c	$[F] = \text{MLT}^{-2}$, $[\alpha] = \text{T}^{-2}$ and their $[I]$ all used $[F] = \text{MLT}^{-2} \neq [I\alpha] = \text{ML}^2\text{T}^{-2}$ so the student's suggestion is incorrect	M1 (AO 1.2) A1 (AO 2.3) [2]	
		d	$[x] = \text{L}$	B1 (AO 2.2a) [1]	
		e	E.g. The relationship could be $Fx = pI\alpha$ for some dimensionless quantity p	B1 (AO 3.5b) [1]	Or: there might be some other term(s) consisting of some dimensionally correct combination of quantities that need to be included
			Total	10	



Question			Answer/Indicative content	Marks	Guidance	
9		a	$\alpha = \beta$ $[u] = LT^{-1}$ or $[v] = LT^{-1}$ $L = L^{\alpha} T^{-\alpha} T^{\gamma}$ or $L^{\alpha} T^{\gamma-\alpha}$ $\alpha = 1$ $\gamma - \alpha = 0$ $\gamma = 1$ and $\beta = 1$	B1 (AO2.2a) B1 (AO3.3) M1 (AO1.1a) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1)	soi – does not need justification Seen No k Could be β or $\beta = 1$ or $\alpha = 1$ if β found	Allow $L = L^{\alpha} T^{-\alpha} T^{\gamma} + L^{\beta} T^{-\beta} T^{\gamma}$ with consistent indices, must be expanded, use BOD
				[6]	<u>Examiner's Comments</u> This question required candidates to use dimensional analysis, a new topic at this level. The question was based around a well-known constant acceleration equation, but required candidates to carry out a complete analysis as if they were not familiar with the equation. Almost all candidates were able to use the correct terms L , T and LT^{-1} for displacement, time and velocity and most recognised that $\alpha = \beta$ for consistency. Very few candidates realised that only one of u and v should be used in the method, since you cannot “add” dimensions. This was not penalised on this occasion. Successful candidates were able to formulate a satisfactory equation. They then used their analysis to compare the indices of L and T in order to determine α , β and γ . Less successful candidates often produced an incorrect analysis, which meant that further marks that depended on it were rarely gained. Some candidates followed up their initial correct analysis with incorrect applications of the rules of indices, with things such as $L^{\alpha} + L^{\beta} = L^{\alpha+\beta}$ or $T^{-\alpha} T^{\gamma} = T^{-\alpha\gamma}$ being seen in working, leading to incorrect answers.	



Question			Answer/Indicative content	Marks	Guidance	
					In this specific question, candidates were able to go straight from the analysis to the solution without further working due to the simplicity of the indices. However, in future examinations, the values may be such that working will be necessary in order to find the correct values.	
		b	If $a = 0$ then $u = v$ and $s = 2kut...$...but "dist = speed $\times$ time" so $k = \frac{1}{2}$	M1 (AO2.1) A1 (AO2.2a) [2]	Must include justification <u>Examiner's Comments</u> This question required candidates to prove that $k = \frac{1}{2}$ from first principles, using the fact that $u = v$ and that $s = ut$ under constant acceleration. A minority of candidates were able to make progress with this item, which was generally accessible only to candidates who had scored full marks in part (a). Some tried to use their incorrect values from part (a) whilst a small number tried to make use of their prior knowledge of the relevant constant acceleration equation, which did not gain any credit.	Do not accept use of prior knowledge of $uvast$
			Total	8		



Question			Answer/Indicative content	Marks	Guidance
10		a	$[P] = T, [m] = M, [h] = L$ $[g] = LT^{-2}$ $M: \alpha = 0$ $T: 1 = -2\gamma \Rightarrow \gamma = -\frac{1}{2}$ $L: 0 = \beta + \gamma \Rightarrow \beta = \frac{1}{2}$	B1(AO3.3) B1(AO3.3) B1(AO1.1) B1(AO3.4) B1(AO1.1) [5]	All three soi
		b	Values of $\frac{P}{\sqrt{h}}$ are: 2.01, 1.37, 2.01 so not consistent	M1(AO2.1) A1(AO2.1) [2]	Or equivalent calculations (or graph plot/sketch)
		c	2.17 must be wrong $1.27 \times \frac{\sqrt{2.50}}{\sqrt{0.40}}$ or $3.81 \times \frac{\sqrt{2.50}}{\sqrt{3.60}}$ So an estimate of the correct value is 3.175	B1(AO3.5a) M1(AO3.4) A1(AO2.2b) [3]	soi oe, eg $k = 6.286...$ or $2.008\sqrt{g}$ Allow 3sf answer 3.17 or 3.18
			Total	10	



Question			Answer/Indicative content	Marks	Guidance	
11		a	$[F] = \text{MLT}^{-2}$ $[\rho] = \text{MLT}^{-2}\text{L}^{-2} = \text{ML}^{-1}\text{T}^{-2}$	B1(AO 2.1) B1(AO 1.1) [2]	AG Must see intermediate step	
		b	$[\rho] = \text{ML}^{-3}$ $[g] = \text{LT}^{-2}$ and $[h] = \text{L}$ $\text{ML}^{-1}\text{T}^{-2} = \text{M}^\alpha \text{L}^{-3\alpha} \text{L}^\beta \text{T}^{-2\beta} \text{L}^\gamma$ $\alpha = 1, \beta = 1$ $-1 = -3\alpha + \beta + \gamma$ $\gamma = 1$	B1(AO 3.3) B1(AO 1.1) M1(AO 1.1a) A1(AO 3.4) M1(AO 1.1) A1(AO 1.1) [6]		
		c	$S = 4.39 \times 10^{-4}$ $h = \text{awrt } 2.0 \text{ m}$	M1(AO 3.4) A1(AO 1.1) [2]	Accept 4.38 or in exact or uncalculated form	Or $\frac{p_2}{p_1} = \frac{h_2}{h_1}$ oe
		d	$[hv^2] = \text{L}^3\text{T}^{-2}$ this is not consistent with the other terms and so the equation cannot be correct	M1(AO 3.3) A1(AO 3.5b) [2]		
			Total	12		



Question			Answer/Indicative content	Marks	Guidance	
12		a	$\rho = ML^{-3}$	B1(AO1.2) [1]	Allow ML^{-3}	
		b	$T = (ML^{-3})^{\alpha} (LT^{-2})^{\beta} (L^2)^{\gamma} = M^{\alpha} L^{-3\alpha+\beta+2\gamma} T^{-2\beta}$ $M: \alpha = 0$ $L: -3\alpha + \beta + 2\gamma = 0$ $T: -2\beta = 1$ $\beta = -\frac{1}{2}$ $\gamma = \frac{1}{4}$	M1(AO3.3) B1(AO1.1) M1(AO3.4) A1(AO1.1) [4]	For $\alpha = 0$ Setting up equations and solving simultaneously to find β and γ For β and γ correct	
		c	$T = C g^{\frac{1}{2}} A^{\frac{1}{4}}$ 2.76	M1(AO1.1) A1(AO3.5c) [2]	Substituting values into their equation for T	$C = 2.755472...$
		d	e.g. According to this model the period is independent of the density e.g. Because $\alpha = 0$	*E1(AO2.2b) dep*E1(AO3.5b) [2]	e.g. It is possible that the period is dependent on another (unknown) quantity e.g. because the value of C may depend on a variable that was not considered	
			Total	9		



Question			Answer/Indicative content	Marks	Guidance	
13			$[G] = \frac{[F][r]^2}{[m_1][m_2]} \text{ or } [G] = \frac{[F][r]^2}{[m]^2}$ $[r] = L \text{ and } [m] = M$ $[F] = M[a] = MLT^{-2}$ So $[G] = M^{-1}L^3T^{-2}$	M1(AO1.1) B1(AO1.1) B1(AO1.1) A1(AO1.1) [4]	Use of rearranged formula to find [G] soi soi Correct answer $M^{-1}L^3T^{-2}$	
			Total	4		



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